

STEGANOGRAPHY FORENSIC DETECTION REPORT

Steganography Detection & Hidden Data Extraction Project
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01 EXECUTIVE SUMMARY

20	10	10	100.0%	100.0%
Total Scanned	Stego Detected	Clean Files	Detection Rate	Overall Accuracy
TRUE POSITIVES	TRUE NEGATIVES	FALSE POSITIVES	FALSE NEGATIVES	
10/10	10/10	0	0	
Stego correctly detected	Clean correctly passed	Clean wrongly flagged	Stego missed	

02 DETECTION METHODOLOGY

#	Method	Description	Applies to
Method 1	Chi-Square Attack	Statistical LSB pair uniformity test.	JPEG / PNG
Method 2	LSB Uniformity Analysis	Bit transition randomness — stego ≈ 50% flips.	ALL
Method 3	Shannon Entropy Analysis	Information content of LSB plane — stego has max entropy.	ALL
Method 4	StegExpose (Java)	Password-independent fusion (Primary Sets + RS Analysis).	JPEG
Method 5	Steghide Extraction	Active password-based extraction. Success = confirmed stego.	JPEG / WAV
Method 6	LSB Extraction (PNG)	Reverses raw LSB embedding without any password.	PNG
Method 7	WAV Audio LSB Analysis	LSB + entropy analysis applied to 16-bit audio samples.	WAV

03 PER-FILE SCAN RESULTS

Filename	Type	Size	Confidence	Extract	Verdict
download (1).jpeg	clean-jpeg	5.9 KB	37.1%	NO	CLEAN
download (2).jpeg	clean-jpeg	5.0 KB	8.1%	NO	CLEAN
download.jpeg	clean-jpeg	11.5 KB	37.6%	NO	CLEAN

Filename	Type	Size	Confidence	Extract	Verdict
images (1).jpeg	clean-jpeg	6.5 KB	36.8%	NO	CLEAN
images.jpeg	clean-jpeg	6.7 KB	30.8%	NO	CLEAN
download (1).png	clean-png	2.7 KB	30.9%	NO	CLEAN
download.png	clean-png	6.2 KB	41.1%	NO	CLEAN
ambient_sample_1.wav	clean-wav	430.7 KB	29.5%	NO	CLEAN
music_sample_1.wav	clean-wav	430.7 KB	30.0%	NO	CLEAN
voice_sample_1.wav	clean-wav	430.7 KB	29.9%	NO	CLEAN
stego_download (1).jpeg	stego-jpeg-steghide	6.4 KB	96.8%	YES	STEGO DETECTED
stego_download (2).jpeg	stego-jpeg-steghide	5.4 KB	68.1%	YES	STEGO DETECTED
stego_download.jpeg	stego-jpeg-steghide	12.2 KB	97.5%	YES	STEGO DETECTED
stego_images (1).jpeg	stego-jpeg-steghide	7.2 KB	96.9%	YES	STEGO DETECTED
stego_images.jpeg	stego-jpeg-steghide	7.1 KB	91.0%	YES	STEGO DETECTED
stego_download (1).png	stego-png-lsb	6.1 KB	100.0%	YES	STEGO DETECTED
stego_download.png	stego-png-lsb	13.9 KB	100.0%	YES	STEGO DETECTED
stego_ambient_sample_1.wav	stego-wav-steghide	430.7 KB	99.6%	YES	STEGO DETECTED
stego_music_sample_1.wav	stego-wav-steghide	430.7 KB	100.0%	YES	STEGO DETECTED
stego_voice_sample_1.wav	stego-wav-steghide	430.7 KB	99.9%	YES	STEGO DETECTED

04 ■■■■ EXTRACTED PAYLOAD PROOF

File	Recovered Payload
stego_download (1).jpeg	[Password: stego123] Secret Message: Server IP = 192.168.10.5 Password = admin123
stego_download (2).jpeg	[Password: stego123] Secret Message: Server IP = 192.168.10.5 Password = admin123
stego_download.jpeg	[Password: stegopass123] Secret Message: Server IP = 192.168.10.5 Password = admin123
stego_images (1).jpeg	[Password: stegopass123] Secret Message: Server IP = 192.168.10.5 Password = admin123
stego_images.jpeg	[Password: stegopass123] Secret Message: Server IP = 192.168.10.5 Password = admin123

stego_download (1).png	Secret Message: Server IP = 192.168.10.5 Password = admin123
stego_download.png	Secret Message: Server IP = 192.168.10.5 Password = admin123
stego_ambient_sample_1.w	[Password: stego123] Secret Message: Server IP = 192.168.10.5 Password = admin123
stego_music_sample_1.wav	[Password: stegopass123] Secret Message: Server IP = 192.168.10.5 Password = admin123
stego_voice_sample_1.wav	[Password: stego123] Secret Message: Server IP = 192.168.10.5 Password = admin123

05 ■■■ RECOMMENDATIONS

■ Deploy batch scanning	Use StegExpose for automated analysis of incoming image datasets.
■ Expand tool coverage	Compile Stegdetect on Linux/WSL to detect JSteg and OutGuess.
■ Visual bit-plane analysis	Use StegSolve to visually inspect LSB planes of suspicious PNGs.
■ Multi-method consensus	No single method is definitive — combine all 7 detection methods.
■ Increase dataset size	Larger datasets reduce Chi-Square false positive rate significantly.
■ Encrypt LSB payloads	Add AES-128 encryption to the PNG LSB embedder for stronger stealth.
■ Credential rotation	Immediately rotate compromised credentials (Server IP 192.168.10.5).

✓ **PROJECT OBJECTIVES ACHIEVED** | Detection Rate: 100.0% | All stego payloads recovered